

Ex ①.

16 μ P $\rightarrow$ data bus width

EEPROM: 256 kB, end: FFFFFFFH, 64 k $\times$ 16b

DRAM: 128 kB, 32 k $\times$ 8b

Step 0: (we need address bus width)

end: FFFFFFH $\rightarrow 5 \times 4 = 20 \rightarrow n = 20$
5 F's

$$n = 20 = 2^{20} \times 1B = 2^{20} B = 1 MB \Rightarrow [0 - (2^n - 1)] = [0 - (2^{20} - 1)]$$

$\downarrow$
nr de
adrese

$$= 0 - 0b \underbrace{10 \dots 0}_{20 \text{ de zero-uri}} - 1$$

$$\rightarrow 0 - 0b \underbrace{1 \dots 1}_{20 \text{ de 1}} \Rightarrow F: 1111 \mid \Rightarrow 20:4=5$$

$\swarrow$ 5 F-uri

$$\Rightarrow [0 - \underline{\text{FFFFFFH}}]$$

Step 1:

• EEPROM: $256 \text{ kB} / (64 \text{ k} \times \underbrace{16b}_{2B}) = 256 \text{ kB} / 128 \text{ kB} = \underbrace{2}_{\text{2 circuits inside a block}}$

$$16:16 = 1 \text{ circuit per block}$$

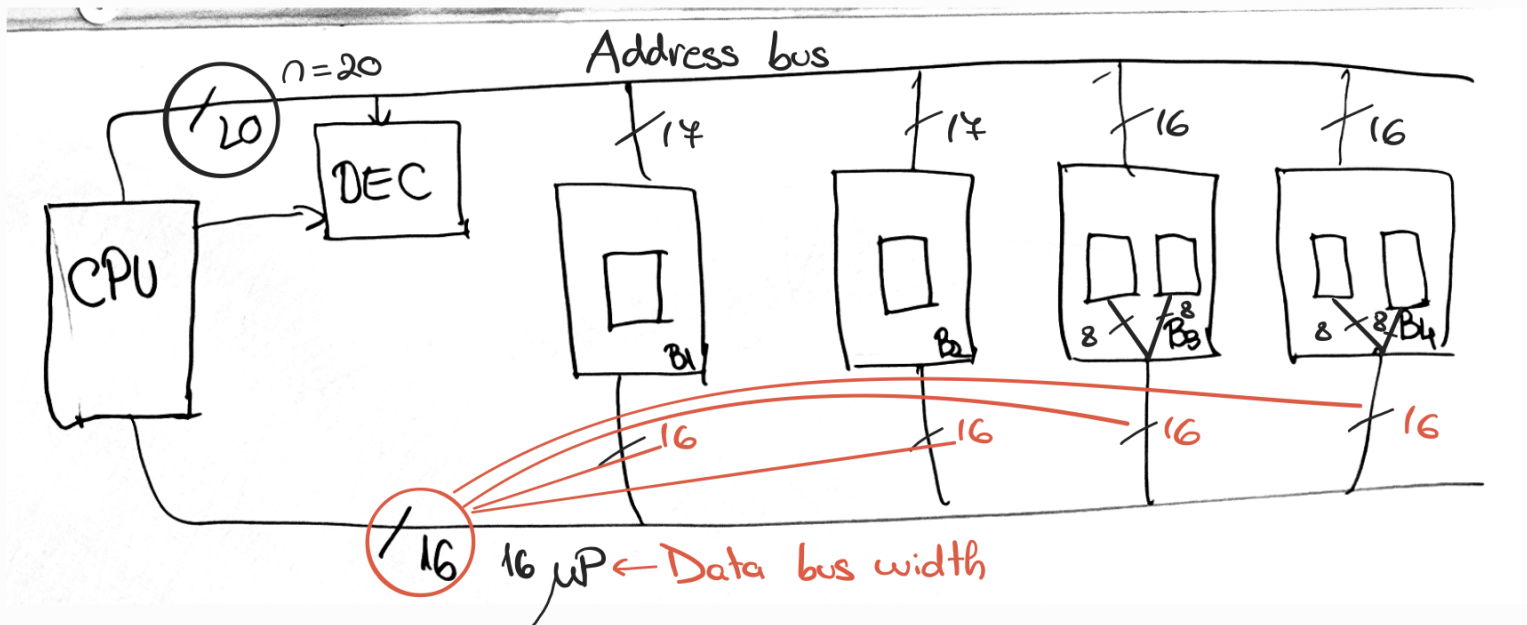
$$2:1 = 2 \text{ blocks: } B_1, B_2$$

• DRAM: $128 \text{ kB} / (32 \text{ k} \times \underbrace{8b}_{1B}) = 128 \text{ kB} / 32 \text{ kB} = \underbrace{4}_{\text{4 circuits inside a block}}$

$$16:8 = 2 \text{ circuits per block}$$

$$4:2 = 2 \text{ blocks: } B_3, B_4$$

Step 2:



Step 3: a.

Block	Number of (n) address lines	Memory size of the block $2^n B$	Address range $[0 \rightarrow (2^n - 1)]$
B_1	$(256 \text{ kB} : 2 = 128 \text{ kB})$ $128 \text{ kB} = 2^{14}$ $\Rightarrow 20000 \text{ H}$	14	$[0000 \ 0000 \text{ H} - 1 \text{ FFFF H}]$
B_2	$(256 \text{ kB} : 2 = 128 \text{ kB})$ $128 \text{ kB} = 2^{14}$ $\Rightarrow 20000 \text{ H}$	14	$[0000 \ 0000 \text{ H} - 1 \text{ FFFF H}]$
B_3	$(128 \text{ kB} : 2 = 64 \text{ kB})$ $64 \text{ kB} = 2^{16}$ $\Rightarrow 10000 \text{ H}$	16	$[0000 \ 0000 \text{ H} - \text{ FFFF H}]$
B_4	$(128 \text{ kB} : 2 = 64 \text{ kB})$ $64 \text{ kB} = 2^{16}$ $\Rightarrow 10000 \text{ H}$	16	$[0000 \ 0000 \text{ H} - \text{ FFFF H}]$

$$2^{14}: \overset{2^1}{1} \overset{2^0}{0} \dots 0 \Rightarrow 20000 \text{ H}, \quad 2^{16}: \overset{2^1}{1} \overset{2^0}{0} \dots 0 \Rightarrow 10000 \text{ H}$$

14 Zero: 16 Zero:

$$2^{14} - 1: 0b \overset{2^1}{1} \overset{2^0}{0} \dots 0 - 1 = 0b \overset{2^1}{1} \dots \overset{2^0}{1} \Rightarrow \text{FFFF H}$$

14 Zero: 16 un-zero:

Step 3: b. Memory Decoder

un interval de
mărimea de sus

Block	Memory Map	
B_1	$[E0000 H - FFFFFFF H]$	(E0000 H is a multiple of 20000 H)
B_2	$[C0000 H - DFFFFFF H]$	(C0000 H is a multiple of 20000 H)
B_3	$[B0000 H - BFFFFFF H]$	(B0000 H is a multiple of 10000 H)
B_4	$[A0000 H - AFFFFFF H]$	(A0000 H is a multiple of 10000 H)

$$\text{Size} = (\text{end} - \text{start} + 1)$$

$$\Rightarrow \text{start} = \text{end} - \text{size} + 1$$

$$\begin{aligned} \Rightarrow & \text{FFFFFF} - \\ & \underline{20000} \\ & (15 - 2 = 13) \Rightarrow D \\ & \underline{DFFFFFF + 1} \\ & E0000 H \end{aligned}$$

$$\begin{aligned} \Rightarrow & DFFFFFF - \\ & \underline{20000} \\ & (13 - 2 = 11) \Rightarrow B \\ & \underline{BFFFFFF + 1} \\ & C0000 H \end{aligned}$$

Step 4: Memory Decoder

a.

Block	A_{19}	A_{18}	A_{17}	A_{16}	A_{15}		A_2	A_1	A_0
B_1	1	1	1	1	1		1	1	1
	1	1	1	0	0		0	0	0
B_2	1	1	0	1	1		1	1	1
	1	1	0	0	0		0	0	0
B_3	1	0	1	1	1		1	1	1
	1	0	1	1	0		0	0	0
B_4	1	0	1	0	1		1	1	1
	1	0	1	0	0		0	0	0

iei pattern-ul cel mai lung (gen pașo' nu se mai repetă)
și apoi tai din cele care nu afectează pattern-ul

$$b. \overline{SEL}_{B_1} = A_{18} \cdot A_{17}$$

$$\overline{SEL}_{B_2} = A_{18} \cdot \overline{A_{17}}$$

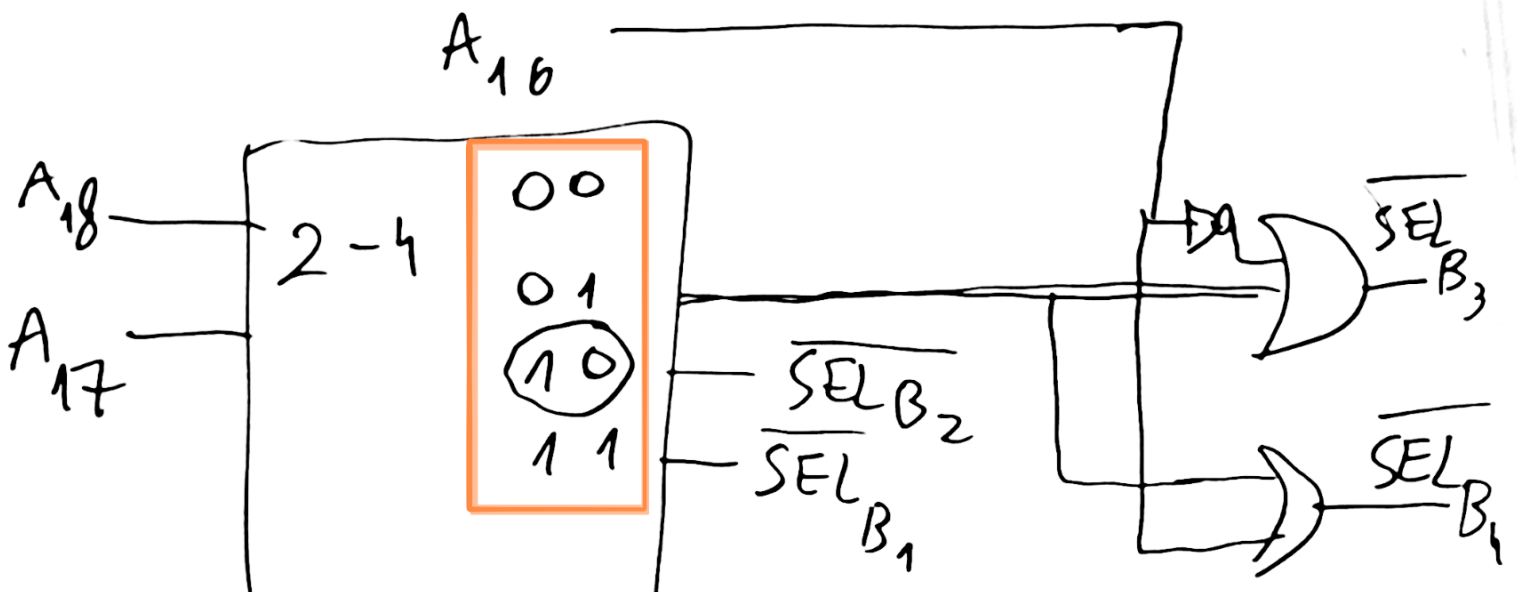
$$\overline{SEL}_{B_3} = (\overline{A_{18} \cdot A_{17}}) \cdot A_{16}$$

this is the extra element

$$= \overline{A_{18}} \cdot A_{17} + A_{16}$$

$$\overline{SEL}_{B_4} = (\overline{A_{18} \cdot A_{17}}) \cdot \overline{A_{16}}$$

$$= \overline{A_{18}} \cdot A_{17} + \overline{A_{16}}$$



Ex ②.

16b μP

EEPROM: 256 kB, start: 40000 H, 64 k $\times$ 16b

DRAM: 256 kB, 64 k $\times$ 1b

Step 0:

$$n = 20 \rightarrow 2^{20} B = 1 MB$$

Step 1:

$$\text{EEPROM: } 256 \text{ kB} / (64 \text{ k} \times \underline{16}) = 256 \text{ kB} / 64 \text{ kB} = 4 \text{ circuits inside the block}$$

$$16 / 8 = 2 \text{ circuits per block}$$

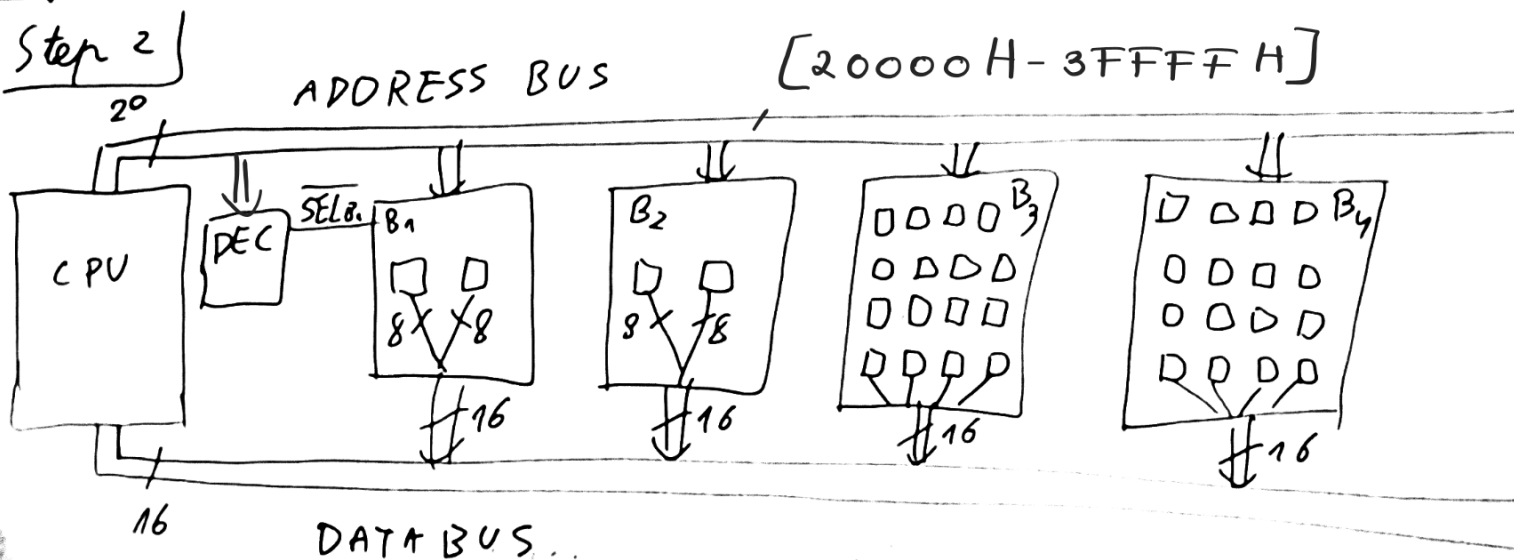
$$4 / 2 = 2 \text{ blocks: } B_1, B_2$$

$$\text{DRAM: } 256 \text{ kB} / (64 \text{ k} \times 1 \text{b}) = \cancel{256 \text{ k}} \times \cancel{8 \text{b}} / \cancel{64 \text{ k}} \times \cancel{1 \text{b}} = 32 \text{ circuits inside the block}$$

$$16 / 1 = 16 \text{ circuits per block}$$

$$32 / 16 = 2 \text{ blocks: } B_3, B_4$$

Step 2:



Step 3:

Block	Number of (n) address lines	Memory size of the block $2^n B$	Address range $[0 \rightarrow (2^n - 1)]$
B_1	$(256 : 2 = 128)$ $128 \text{ kB} = 2^{17} B$ $\Rightarrow 20000H$	17	$[00000000H - 1FFFFFFH]$
B_2	$(256 : 2 = 128)$ $128 \text{ kB} = 2^{17} B$ $\Rightarrow 20000H$	17	$[00000000H - 1FFFFFFH]$
B_3	$(256 : 2 = 128)$ $128 \text{ kB} = 2^{17} B$ $\Rightarrow 20000H$	17	$[00000000H - 1FFFFFFH]$
B_4	$(256 : 2 = 128)$ $128 \text{ kB} = 2^{17} B$ $\Rightarrow 20000H$	17	$[00000000H - 1FFFFFFH]$

Step 3: b. Memory Decoder

Block	Memory Map	
B_1	$[40000H - 5FFFFFFH]$	$(40000H + 1FFFFFFH) = 5FFFFFFH$
B_2	$[60000H - 7FFFFFFH]$	$(60000H + 1FFFFFFH) = 7FFFFFFH$
B_3	$[20000H - 3FFFFFFH]$	$(20000H + 1FFFFFFH) = 3FFFFFFH$
B_4	$[00000H - 1FFFFFFH]$	$(00000H + 1FFFFFFH) = 1FFFFFFH$

Step 4: Memory Decoder

a.

Block	A_{19}	A_{18}	A_{17}	A_{16}	A_{15}	...	A_2	A_1	A_0
B_1	end 0	1	0	1	1	...	1	1	1
	start 0	1	0	0	0	...	0	0	0
B_2	end 0	1	1	1	1	...	1	1	1
	start 0	1	1	0	0	...	0	0	0
B_3	end 0	0	1	1	1	...	1	1	1
	start 0	0	1	0	0	...	0	0	0
B_4	end 0	0	0	1	1	...	1	1	1
	start 0	0	0	0	0	...	0	0	0

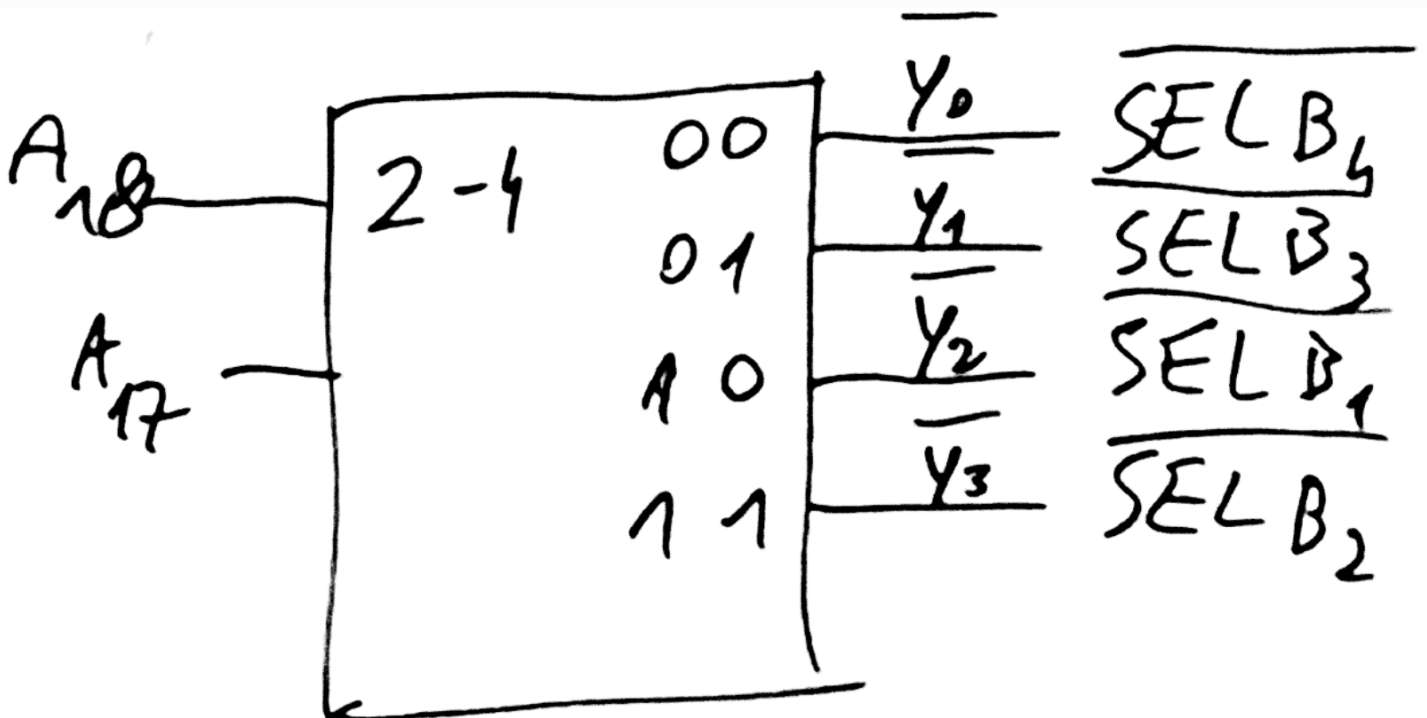
incomplete decoder: because we deleted A_{19}
complete decoder: we delete nothing

b. $\overline{SEL}_{B_1} = A_{18} \cdot \overline{A_{17}}$

$\overline{SEL}_{B_3} = \overline{A_{18}} \cdot A_{17}$

$\overline{SEL}_{B_2} = A_{18} \cdot A_{17}$

$\overline{SEL}_{B_4} = \overline{A_{18}} \cdot \overline{A_{17}}$



16μ P n=20

- 96KB EEPROM ending at FFFFFH, 16K×16b

- 128KB DRAM, 64K×2b

Step 0) $2^{20} = 1 \text{ MB}$

Step 1)

- EEPROM: $96\text{KB} / (16\text{K} \times \frac{16\text{b}}{2\text{B}}) = 96\text{KB} / 32\text{KB} = 3 \text{ circuits}$

$16/16 = 1 \text{ c/blck} \Rightarrow$

- DRAM: $128\text{KB} / (64\text{K} \times 2\text{b}) = \frac{3/1 = 3 \text{ blocks } B_1, B_2, B_3}{\frac{128\text{K} \times 8\text{b}}{64\text{K} \times 2\text{b}}} = 8 \text{ circuits}$

$16/2 = 8 \text{ c/blck} \Rightarrow$

$8/8 = 1 \text{ blk } B_4$

Step 2

size = end - start + 1

end = size + start - 1

start = end - (size - 1)

Step 3)

Block	n	2^n	$[0 \rightarrow (2^n - 1)]$	Memory Map
B ₁	15	32KB = 2^{15} 8000H	[00000H → 07FFFH]	[F8000H - FFFFFFFH] ⇒
B ₂	15	8000H	[00000H → 07FFFH]	[F0000H - F7FFFFH]
B ₃	15	8000H	[00000H → 07FFFH]	[E8000H - EFFFFFFH]
B ₄	17	128KB = 2^{17} 20000H	[00000H → 1FFFFH]	[00000H - 1FFFFH]

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Hola

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Say Dora

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★ Dora the Explora ★

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Step 4

a)

Block	A_{19}	A_{18}	A_{17}	A_{16}	A_{15}	...	A_2	A_1	A_0
B_1	1	1	1	1	1	...	1	1	1
	1	1	1	1	1	...	0	0	0
B_2	1	1	1	1	0	...	1	1	1
	1	1	1	1	0	...	0	0	0
B_3	1	1	1	0	1	...	1	1	1
	1	1	1	0	1	...	0	0	0
B_4	0	0	0	1	1	...	0	0	0
	0	0	0	1	1	...	1	1	1

b)

$$\overline{SEL_{B_1}} = \overline{(A_{19} \cdot A_{18} \cdot A_{17}) \cdot A_{16} \cdot A_{15}} = \overline{A_{19} \cdot A_{18} \cdot A_{17} \cdot A_{16}} + \overline{A_{15}}$$

$$\overline{SEL_{B_2}} = \overline{A_{19} \cdot A_{18} \cdot A_{17} \cdot A_{16} \cdot \overline{A_{15}}} = \overline{A_{19} \cdot A_{18} \cdot A_{17} \cdot A_{16}} + A_{15}$$

$$\overline{SEL_{B_3}} = \overline{A_{19} \cdot A_{18} \cdot A_{17} \cdot \overline{A_{16}} \cdot A_{15}} = \overline{A_{19} \cdot A_{18} \cdot A_{17} \cdot A_{16}} + A_{15}$$

$$\overline{SEL_{B_4}} = \overline{\overline{A_{19}} \cdot \overline{A_{18}} \cdot \overline{A_{17}} \cdot (\overline{A_{16}} + A_{16})}$$

$$= \overline{\overline{A_{19}} \cdot \overline{A_{18}} \cdot \overline{A_{17}} \cdot \overline{A_{16}}} + \overline{\overline{A_{19}} \cdot \overline{A_{18}} \cdot \overline{A_{17}} \cdot A_{16}}$$

